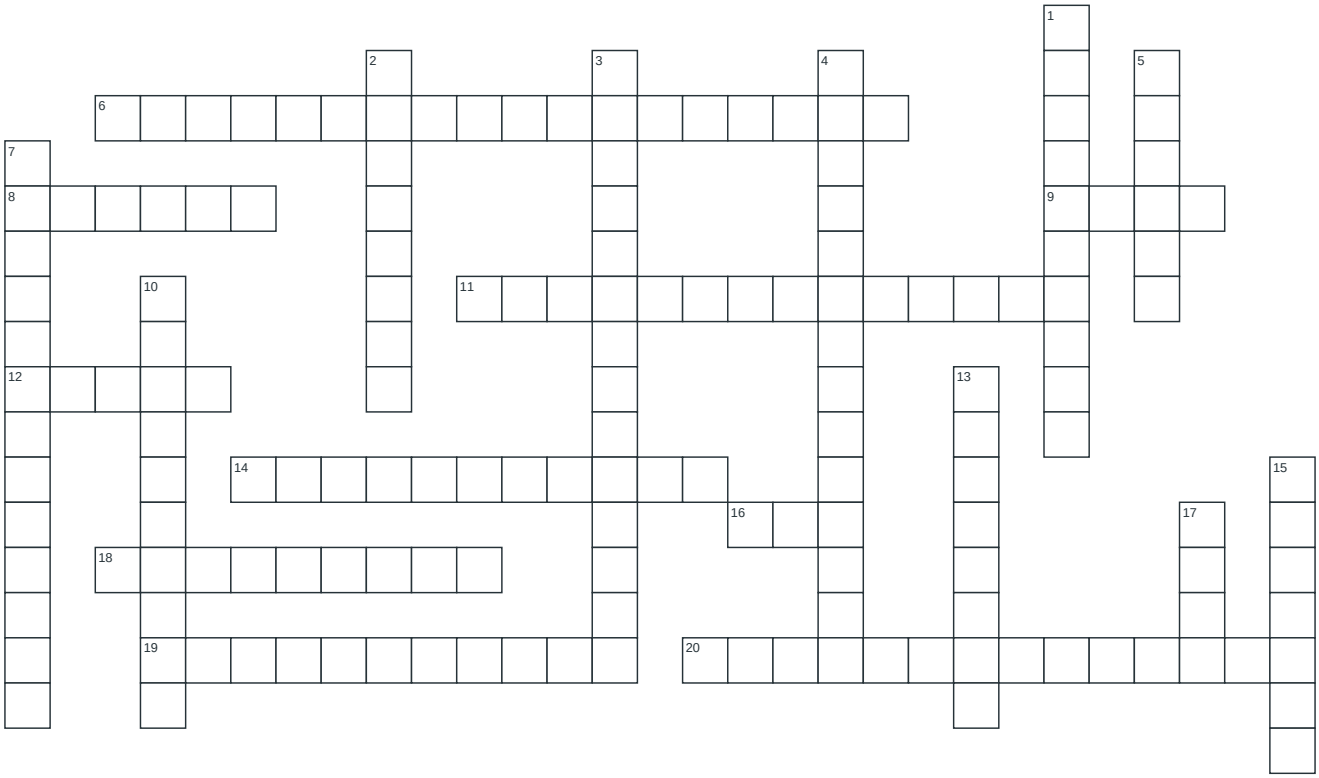
# DNA RNA and Protein Synthesis Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



## ACROSS

**6.** Process by which cells develop structures suited to particular functions.
**8.** Five-carbon sugar found in RNA
**9.** One DNA strand reads ACGT. Which bases pair directly across from it
**11.** Plant-cell compartment that stores water and helps maintain pressure.
**12.** Three-nucleotide sequence in messenger RNA that specifies an amino acid or stop
**14.** Organelle where photosynthesis occurs in plants and algae.
**16.** Molecule that stores hereditary information in most cells
**18.** Cell contents outside the nucleus and inside the plasma membrane.
**19.** Use of an RNA message to assemble a polypeptide
**20.** Boundary that regulates exchange between a cell and its surroundings.

## DOWN

**1.** States that living things are made of cells and that cells come from existing cells.
**2.** Rigid outer layer that supports many plant, fungal, and bacterial cells.
**3.** A supplied key says GCU and GCC both encode alanine. What follows from changing GCU to GCC
**4.** Organelle that modifies, sorts, and packages many proteins.
**5.** RNA base that generally replaces thymine
**7.** Synthesis of RNA from a DNA template
**10.** Cell type that lacks a membrane-bound nucleus.
**13.** Cell structure that assembles proteins.
**15.** Membrane-bound organelle that contains most eukaryotic DNA.
**17.** RNA molecule that carries an amino acid to a ribosome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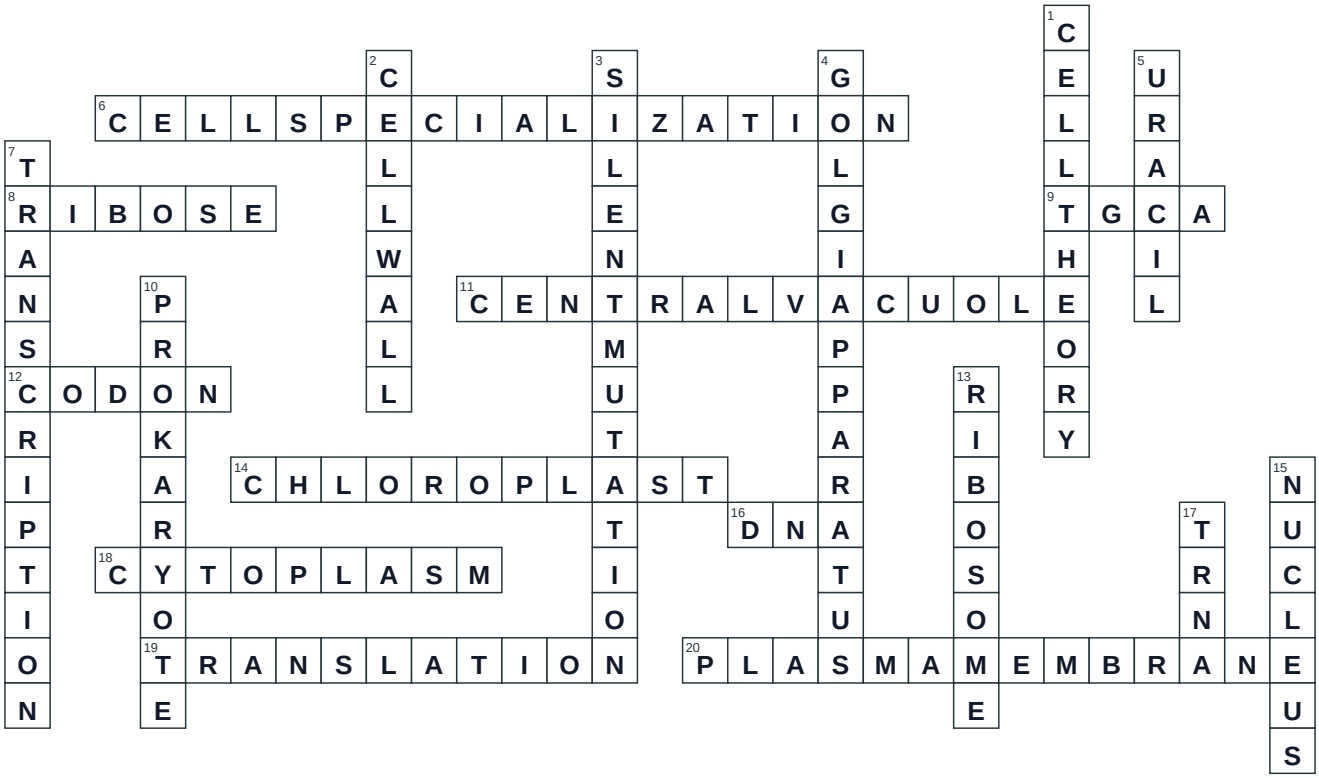
# DNA RNA and Protein Synthesis Crossword

Hard | 20 clues | Teacher Answer Key

### TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.




## ACROSS

**6.** Cell specialization
**8.** Ribose
**9.** TGCA
**11.** Central vacuole
**12.** Codon
**14.** Chloroplast
**16.** DNA
**18.** Cytoplasm
**19.** Translation
**20.** Plasma membrane

## DOWN

**1.** Cell theory
**2.** Cell wall
**3.** Silent Mutation
**4.** Golgi apparatus
**5.** Uracil
**7.** Transcription
**10.** Prokaryote
**13.** Ribosome
**15.** Nucleus
**17.** tRNA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